# Segmentation & Representation Benchmark Matrix

Consolidated validation across three corpora (MeerKAT, LibriSpeech, InfantMarmosetsVox), empirical


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## Terminology — what the pipeline discovers in animal audio

Speech word-discovery methods (DUSTED, ES-KMeans, Bayesian word segmentation, SSL-unit clustering

| | |
|---|---|
| speech (human, has GT) | animal (no GT meaning) |
| word | recurrent acoustic term / motif |
| word-like unit | motif-like acoustic term |
| phoneme / subword | sub-call acoustic unit |
| lexicon | motif inventory |
| sentence | bout / vocalisation sequence |

All calltype/caller "enrichment" reported below is **acoustic/structural association, not semantics

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## 1. Event-detection F1 (which segmenter, which regime)

| corpus (regime) | method | metric | F1 | P | R | ref |
|---|---|---|---|---|---|---|
| **MeerKAT** (callsign noise, 8 kHz) | raw NBD | onset 1-to-1 | 0.069 | — | — | massive over-segmentation |
| | sweep-best threshold | onset 1-to-1 | 0.165 | — | — | thresholds insufficient |
| | v2 scalar GBT (19 feat) | onset 1-to-1 | ~0.26 | — | — | tabular baseline (frozen) |
| | v3/v4 post-onset GBT | onset 1-to-1 | ~0.21 | — | — | tabular **ceiling** |
| | **PCEN+CRNN** | onset 1-to-1 | **0.703** | — | — | **8.9× raw NBD → gap closed** |
| **LibriSpeech** (scalar GBT speech, word-level) | scalar GBT speech, word-level | word 1-to-1 / combined | 0.450 / 0.638 | — | — | CPC-free |
| | CPC-context GBT (+dim) | word 1-to-1 / combined | 0.458 / 0.794 | — | — | CPC adds +0.156 combined |
| | **NBD+CPC calibrated** | **combined (§1 metric)** | **0.836** | 0.77 | 0.92 | **speech winner** |
| | PCEN+CRNN (retrained) | word 1-to-1 / combined | 0.590 / 0.652 | 0.80 | 0.47 | CRNN is the **speech bottleneck** |
| **InfantMarmosetsVox** (clean, adult) | **PCEN+CRNN (retrained)** | onset 1-to-1 | **0.875** | **0.982** | 0.789 | **strongest result** |

**Router read:** PCEN+CRNN wins for discrete calls in noise (0.70 meerkat → 0.88 marmoset); NBD+CPC

### Tolerance sweep (apples-to-apples — `scripts/tolerance_sweep.py`)

The headline F1s used different tolerances (MeerKAT 100ms, marmoset 200ms). At **matched**

tolerance

| onset F1 | 50ms | 100ms | 200ms |
|---|---|---|---|
| MeerKAT (500 clips) | 0.662 | 0.707 | 0.724 |
| InfantMarmosetsVox (30 test) | **0.864** | **0.865** | **0.867** |

| offset F1 | 50ms | 100ms | 200ms |
|---|---|---|---|
| MeerKAT | 0.549 | 0.664 | 0.702 |
| InfantMarmosetsVox | **0.785** | **0.860** | **0.866** |

Marmoset onset F1 is tolerance-*insensitive* (0.864→0.867) because precision ≈ 0.98 (near-

zero false

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## 2. HECB internal-boundary mechanism (oracle-parent, no CRNN)

The internal mechanism decoupled from the detector — given valid parent events, does HECB

recover m

| corpus | parent events | sub-event recovered | F1 | P | R |
|---|---|---|---|---|---|
| **LibriSpeech** | oracle word spans | in-word phoneme start | **0.636** | 0.781 | 0.537 |
| LibriSpeech | CRNN word events | in-word phoneme start | 0.409 | 0.516 | 0.339 |

**Read:** the mechanism works at 0.636 with good parents but degrades to 0.41 with poor

(CRNN) pa

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## 3. Whole / sub-call / pooled representation recovery

The core claim: sub-call features **augment** whole-event representations when pooled, but

are weake

### Call-type recovery

| corpus | split | whole V | pooled V | whole probe | pooled probe | whole macroF1 | pooled macroF1 |
|---|---|---|---|---|---|---|---|
| **MeerKAT** (end-to-end, 1500 clips) | — | 0.246 | **0.274** | 0.611 | **0.633** | — | — |
| **InfantMarmosetsVox** (end-to-end) | — | 0.344 | **0.379** | 0.934 | **0.940** | — | — |
| InfantMarmosetsVox (oracle) | random | 0.431 | 0.416 | 0.935 | 0.932 | 0.487 | **0.566** |
| InfantMarmosetsVox (oracle) | within-callid | 0.477 | 0.475 | 0.944 | **0.947** | 0.557 | **0.582** |

*Sub-call alone is consistently weaker on V (loses whole-call context) but highest on probe

(sub-fragm

### Caller-ID recovery (InfantMarmosetsVox oracle, random split, 10 callers) — **the novel finding**

| representation | V | NMI | probe | macro-F1 |
|---|---|---|---|---|
| whole call | 0.178 | 0.070 | 0.459 | 0.364 |
| **whole + sub-call pool** | **0.197** | **0.083** | **0.477** | **0.392** |
| sub-call | 0.115 | 0.029 | 0.317 | — |

**HECB internal boundaries recover caller identity — pooled wins on all four metrics.**

Individual

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## 4. The pooled-adds-value thesis (cross-corpus summary)

| corpus | label | pooled > whole? | on which metric |
|---|---|---|---|
| MeerKAT | call-type | | V, NMI, probe (clean) |
| InfantMarmosetsVox (end-to-end) | call-type | | V, NMI, probe |
| InfantMarmosetsVox (oracle) | call-type | mixed | macro-F1 (+0.079); whole edges V |
| InfantMarmosetsVox (oracle) | **caller ID** | | **all 4 metrics** |

**Refined conclusion (post-controls):** Sub-call structure augments whole-call

represent

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## 5. Segmentation router (canonical)

| regime | method | HECB role | example |
|---|---|---|---|
| discrete calls in noise | PCEN+CRNN | internal annotator | meerkat (0.70), marmoset (0.88) |
| continuous speech-like | NBD+CPC calibrator | optional | LibriSpeech (0.836 combined) |
| exploratory / unknown | NBD + ABSTAIN | none | new species (conservative) |

**Updated core claim (project thesis):** The pipeline uses a regime-specific segmentation

router: PC

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## Caveats

All F1s are one-to-one greedy matching (no duplicate-TP inflation). MeerKAT onset

toleranc

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## 6. Cross-species 4-way ablation (raw-NBD oracle, 2026-06-22)

`scripts/cross_species_ablation.py` — per-file oracle-parent HECB (raw NBD internal

boundarie

| corpus (kHz) | label | whole V | pooled V | **sequence V** | whole probe | pooled probe | seq probe |
|---|---|---|---|---|---|---|---|
| **adult marmoset (vocalization, 96) | call-type (6) | 0.458 | 0.434 | **0.468** | 0.824 | 0.832 | **0.837** |
| **egyptian fruit bat (250) | Context (2) | 0.152 | 0.152 | 0.153 | 0.547 | 0.556 | 0.543 |
| egyptian fruit bat (250) | Emitter (ID) | 0.291 | 0.291 | 0.291 | 0.211 | 0.191 | 0.195 |

**Honest read — this REFINES the pooled-adds-value thesis:**

**The re

**Revised cross-species claim:** sub-call structure adds value where (a) the internal

boundarie

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## 7. Typed HECB vs raw NBD on adult marmoset (boundary-quality test, 2026-06-22)

`scripts/typed_hecb_vs_raw.py` — same eval files (2491 calls, 6 types), two internal-

boundary
lacked.

| method | whole V | pooled V | pooled−whole | pooled probe | sub-call V | sub-call probe |
|---|---|---|---|---|---|---|
| raw NBD | 0.450 | 0.418 | −0.032 | 0.800 | 0.173 | 0.487 |
| **typed HECB** | 0.450 | **0.431** | **−0.019** | **0.814** | **0.255** | **0.527** |

**Honest result:** typed HECB improves sub-call fragment quality (V 0.255 vs 0.173, probe

0.527 vs (

**Cross-experiment pattern (now consistent):** the sub-call advantage appears on

**supervis

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## 8. Caller-ID confound controls (InfantMarmosetsVox, 2026-06-22)

`scripts/caller_id_controls.py` — validates the novel caller-ID finding against confounds.

2828 vocs

### Baseline + within-call-type (V / NMI / probe)

| condition | whole V | pooled V | pooled−whole V | whole probe | pooled probe |
|---|---|---|---|---|---|
| **all call types** | 0.185 | **0.203** | +0.018 | 0.452 | **0.479** |
| within Phee (10 callers) | 0.223 | **0.234** | +0.011 | **0.529** | 0.465 |

| | | | | | |
|---|---|---|---|---|---|
| within Twitter (10 callers) | 0.251 | **0.261** | +0.010 | 0.594 | **0.601** |

### Trivial baselines (probe accuracy)

| baseline | probe acc | macro-F1 | interpretation |
|---|---|---|---|
| duration-only | 0.165 | 0.070 | near-chance (0.10) — **NOT duration** |
| calltype-onehot | 0.138 | 0.059 | near-chance — **NOT call-type confound** |
| whole-fp (dur-regressed) | 0.435 | 0.393 | survives duration removal |

**Verdict:** the caller-ID finding is **genuine** — it survives all confound controls.

The poole

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## 9. Track C bat context ablation (acoustic + social + behavioral, 2026-06-22)

`scripts/bat_trackc_ablation.py` — the scientific gap: does combining modalities recover

context b

### Context recovery — the A–G pairwise ablation matrix (the key target)

| model | features | probe acc | macro-F1 | lift |
|---|---|---|---|---|
| **A** | acoustic-only (fingerprint) | 0.249 | 0.218 | 2.7× |
| **B** | social-only (emitter+addressee) | 0.267 | 0.223 | 2.9× |
| **C** | behavior-only (pre/post actions) | 0.087 | 0.162 | 1.0× |
| **D** | acoustic + social | **0.393** | **0.341** | **4.3×** |
| **E** | acoustic + behavior | 0.298 | 0.307 | 3.3× |
| **F** | social + behavior | 0.369 | 0.327 | 4.1× |
| **G** | acoustic + social + behavior | **0.424** | **0.355** | **4.7×** |

**The pairwise decomposition reveals where the synergy lives:**

**D (aco

### Leakage / generalization controls (reviewer-proof)

| split | A acoustic | B social | D acous+social | G full |
|---|---|---|---|---|
| **random** | 2.7× | 2.9× | **4.3×** | **4.7×** |
| **social-shuffle** (emit/addr permuted) | 2.7× | **1.0×** | **3.2×** | 4.0× |

| | | | | |
|---|---|---|---|---|
| **emitter-held-out** (GroupShuffleSplit) | 2.6× | 2.8× | **4.1×** | **4.8×** |

**Social-shuffle** collapses B to chance (1.0×) and drops D by 1.1× (4.3→3.2) → ~69% of D's lift o

### Manuscript-ready Track C conclusion

Acoustics encode call form and caller-associated structure, but ethological context requires *

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## 10. Sub-call motif structure report (InfantMarmosetsVox, 2026-06-22)

`scripts/subcall_motifs.py` + `scripts/motif_controls.py` — 2423 vocalisations, k=12 fingerprin

### Position-specialized clusters (onset/body/offset roles)

| cluster | first% | middle% | last% | **role** | calltype V | caller V |
|---|---|---|---|---|---|---|
| **7** | **84.0** | 15.7 | 0.2 | **onset** | — | — |
| **1** | 0.6 | 4.2 | **92.4** | **offset** | — | — |
| **0** | 2.3 | **97.6** | 0.1 | **body** | — | — |
| **10** | 0.0 | **100.0** | 0.0 | **body** | — | — |
| **11** | 0.2 | **99.8** | 0.0 | **body** | — | — |
| **3** | 29.7 | **68.3** | 2.0 | body (onset-leaning) | — | — |

### Calltype-enriched motifs (top by V)

| level | motif | count | calltype V | p |
|---|---|---|---|---|
| 1-gram | cluster 0 | 1408 | 0.653 | <0.0001 |
| 1-gram | cluster 7 | 1092 | 0.650 | <0.0001 |
| 2-gram | (0, 0) | 462 | 0.398 | <0.0001 |
| 2-gram | (0, 7) | 307 | 0.372 | <0.0001 |
| 3-gram | (11, 11, 11) | 175 | 0.242 | <0.0001 |
| 3-gram | (0, 7, 0) | 104 | 0.227 | <0.0001 |

### Control summary

| control | result |
|---|---|
| bootstrap stability (bigram Jaccard) | **1.000** (STABLE) |
| bootstrap stability (trigram Jaccard) | **0.773** (STABLE) |
| position enrichment | onset/body/offset roles confirmed (84%/9 |
| within-Phee caller motifs | (10,0) V=0.186 p<0.0001 — caller signal |
| bigram-preserving trigram null | 374 vs 381±5 (5th pct) — trigrams ≈ bigr |
| token-shuffle bigram richness | 94 vs 104±1 — **non-random** (constraine |

### Manuscript-ready claim

HECB-derived sub-call units are stable, non-random, and positionally specialized.

Cluster 7

## 11. DUSTED-style recurrent-subsequence motif discovery (3-way method comparison, 2026-06-22)

Adapts spoken-term-discovery (DUSTED) to **recurrent acoustic terms** in animal audio

(`diagnos

**exact n-grams** — windows match iff cluster-ID tuples are identical (fragments every

substitut

Run: `python scripts/hecb_dusted_marmoset.py --n-eval 1500` (`scripts/hecb_dusted_marmoset.py`).

### Method comparison (the decisive question — variant tier, DTW thr=0.35)

| method | #motifs | total support | coverage | mean calltype V | mean caller V |
|---|---|---|---|---|---|
| exact n-grams | 376 | 6236 | 0.155 | 0.109 | 0.102 |
| fixed-length cosine | 18 | 5988 | 0.509 | 0.289 | 0.125 |
| **DTW with indels** | **2** | **6970** | **0.516** | **0.508** | **0.170** |

DTW merges variants into the fewest, richest families: highest support/coverage and

**calltype

### Honest caveats (do not overclaim)

1. **Degenerate inventory.** InfantMarmosetsVox sub-call structure is dominated by ONE

recurren

2. **Order-shuffle null is conservative for repeats.** The within-call order shuffle

preserv

3. **Bootstrap Jaccard low (0.16–0.18).** With a degenerate inventory (1 dominant + sparse minors)

### Manuscript-ready claim

DTW-with-indels is the right subsequence-matching primitive for recurrent-acoustic-term discovery

## 12. Bengalese-finch song — the rich-repertory positive control (GT-validated, 2026-06-22)

The marmoset caveat (§11) predicted a richer inventory from a richer corpus. Bengalese finch song

### 3-way method comparison (mean across Bird0/1/2)

| method | #motifs | coverage | label purity (mean / wmean) |
|---|---|---|---|
| exact n-grams | 106 | 0.84 | 1.00 / 1.00 |
| fixed-length cosine | 104 | 0.90 | 0.97 / **0.895** |
| DTW with indels | 81 | 0.90 | 0.83 / 0.59 |

The expected, clean method trend — and crucially, **fixed-length cosine recovers the GT syllable m

### DTW recovers real syllable motifs blind from acoustics (selected, per bird)

| bird | motif (GT labels) | support | purity | note |
|---|---|---|---|---|
| Bird1 | `[1,0]` | 304 | **0.90** | clean 2-syllable motif across 141 cal |
| Bird2 | `[4,5]` | 158 | **1.00** | perfectly recovered bigram, 118 calls |
| Bird2 | `[5,6]` | 148 | 0.78 | — |
| Bird2 | `[10,11,12,13]` | 209 | 0.51 | 4-syllable motif |
| Bird0 | `[2,3,4]` / `[6,2,3]` / `[5,6,15]` | 6–10 | 0.51–0.56 | trigrams |

### Nulls + stability (per bird)

Observed DTW support is **2.0–2.3× the order-shuffle null** on every bird (p=0.111 is the

n_perm=8

### Manuscript-ready claim

On Bengalese-finch song — the right dataset class for diverse recurrent acoustic terms —

DUSTED c

## 13. Egyptian fruit bat — emitter/social context; graded, not discrete (2026-06-22)

The stress test the Bengalese result (§12) set up: does a *socially annotated* corpus

form recu

### The decisive finding: Emitter V ≫ Context V (every method)

| method | #fams | coverage | H(norm) | top-frac | **Emitter V** | **Context V** |
|---|---|---|---|---|---|---|
| exact n-grams | 582 | 0.29 | 0.91 | 0.04 | 0.463 | 0.164 |
| fixed-length cosine | 155 | 0.78 | 0.60 | 0.43 | 0.454 | 0.167 |
| DTW with indels | 105 | 0.82 | 0.32 | 0.72 | 0.467 | 0.163 |

**Emitter V ≈ 0.45–0.47 vs Context V ≈ 0.16–0.17 (~2.8×) — rock-solid across all three

methods.*

### Bat is graded/intermediate, not discrete-syllable like Bengalese

The diverse-vs-degenerate diagnostic (the user's `family_entropy` / `top_family_support_fraction`):

bat sits **

### Honest caveats

**Null modest** (obs/null ≈ 1.23×, p=0.2 at n_perm=4): the dominant `[12,12]` repeat is

composi

### Manuscript-ready claim

On Egyptian fruit bat, recurrent acoustic-term families are tied to **emitter/caller

(0.45–0.4

## 14. Unsupervised word-structure benchmark — HECB vs SSL discrete units (LibriSpeech, 2026-06-22)

The decisive external comparison: is HECB's boundary-aware internal sub-unit sequence

competiti

### Headline table

| unit source | within | across | a_dur | a_phn | ratio | NN acc | NN_xspk | motif→word V |
|---|---|---|---|---|---|---|---|---|
| HECB | 0.908 | 0.930 | 0.939 | 0.929 | 0.976 | 0.089 | 0.085 | 0.326 |
| HuBERT-km(50) | 0.928 | 0.920 | 0.924 | 0.923 | **1.008** | 0.052 | 0.052 | 0.302 |
| **CPC-km(50)** | 0.900 | 0.943 | 0.944 | 0.943 | **0.954** | **0.191** | 0.178 | 0.322 |
| **fingerprint(328)** | 0.897 | 0.931 | 0.934 | 0.933 | 0.964 | 0.147 | 0.142 | **0.347** |

### Controls (all clean)

`a_dur` / `a_phn` (across-word baseline matched on duration / phoneme-count) ≈ `across`

for every

### Findings

1. **CPC-km wins** raw same-word consistency (ratio 0.954, NN 0.191). The pipeline's own

SSL mo

### Honest caveats (do not overclaim)

**HuBERT's failure is metric-specific**, not a defeat of HuBERT as a representation.

The zero

### Manuscript-ready claim

On a controlled LibriSpeech same-word-consistency benchmark, HECB's boundary-aware

sub-unit s

> **Fingerprint note (all experiments):** the `fingerprint` source is `afp.extract(...).combined`

> = the
species-n

## 15. Stage C + D — DUSTED term discovery & ES-KMeans clustering (HuBERT under DTW; LibriSpeech, 2026-06-23)

Two follow-on stages closing the §14 "HuBERT fails" loop (was it the metric, not the unit?).

Both reus

### Stage C — DUSTED word-term discovery (DTW, HuBERT's native metric; `dusted_librispeech_words.py`)

Each word instance = one CallUnits (discrete-unit sequence, `calltype=word`). DUSTED module run

per source

| unit source | exact word_V | DTW word_V | DTW fams | DTW cov | top-frac |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| HECB(k32) | 0.409 | 0.425 | 254 | 0.048 | 0.02 |
| HuBERT-km(50) | 0.415 | 0.418 | 501 | 0.122 | 0.02 |
| CPC-km(50) | 0.418 | 0.431 | 651 | 0.075 | 0.01 |
| **fingerprint(32)** | **0.448** | **0.458** | 259 | 0.078 | 0.03 |

**HuBERT recovers under DTW**: word_V 0.418 (vs §14's Levenshtein *failure* ratio 1.008). So

§14's "Hu

### Stage D — whole-word embedding clustering (ES-KMeans lexicon core; `eskmeans_librispeech.py`)

Oracle GT word spans → whole-word acoustic embedding → k-means (k=60) → evaluate vs GT word

labels. Co

| embedding | V-measure | ARI | purity | token-F1 |
|---|---|---|---|---|
| **fingerprint (44-D)** | **0.481** | 0.061 | **0.276** | **0.276** |
| CPC-mean (256-D) | 0.436 | 0.036 | 0.242 | 0.242 |
| HECB-pooled (44-D) | 0.392 | 0.012 | 0.198 | 0.198 |
| HuBERT-mean (768-D) | 0.338 | −0.003 | 0.172 | 0.172 |

**Fingerprint clusters words best** (V=0.481, token-F1 0.276 = 1.6× chance over 88 types);

HuBERT-m

### Manuscript-ready claim

Across four complementary LibriSpeech word-structure tests (§14 Levenshtein, Stage C DTW term

discovery

## 16. HuBERT sanity ablation — the conclusion is locked (2026-06-23)

The reviewer-safe sweep: does ANY HuBERT configuration (layer × k × collapse × pooling)

beat the f

### Track 1 — sequence consistency (18 configs: layer ∈ {6,9,12} × k ∈ {50,100,200} × {col,raw})

Every single HuBERT config has ratio **1.001–1.018** (within ≥ across: *no* same-word structure)

### Track 2 — whole-word clustering (6 configs: layer ∈ {6,9,12} × pool ∈ {mean,stat})

Best HuBERT: **L6 stat (V=0.331, tokF1=0.207, ARI=−0.006)**. Fingerprint REF: V=**0.517**, tokF1=**(

### Locked conclusion

Across **24 HuBERT configurations** spanning the standard zero-resource design space (layers 6/9/12, k

## 17. Bat held-out emitter validation — the claim generalizes (2026-06-23)

The §13 claim (bat recurrent acoustic-term families track emitter identity ~2.8× more than context) v

| metric | Train (70%) | Held-out (30%) |
|---|---|---|
| Emitter V | 0.221 | **0.375** |
| Context V | 0.085 | 0.149 |
| **ratio** | 2.60× | **2.52×** |

85.5% of held-out windows assigned; 98.5% bout coverage. Held-out assignment is **diverse** (entropy 

## 18. RookID individual-ID — whole-call fingerprint wins, extends bat emitter finding (2026-06-23)

The headline RookID experiment: does hierarchical sub-call structure improve individual identificat

| representation | Source V | NMI | macro-F1 | #train | #test | #src |
|---|---|---|---|---|---|---|
| **whole-call fp (44-D)** | **0.384** | **0.320** | **0.287** | 2055 | 944 | 14 |
| HECB-pooled fp (44-D) | 0.321 | 0.242 | 0.263 | 2055 | 944 | 14 |

**Findings:**

1. Whole-

**Dataset summary**: 15,008 calls (excl Inc/Pls), 15 named rooks, 514 events (many "sing"

variants –

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## 19. Clusterer robustness of the pooled representation — k-means is NOT the confound (2026-06-23)

`scripts/marmoset_clusterfn_compare.py` — answers a question raised in §6/§7: is the weak

pooled-vs

| clusterer | pooled V | pooled NMI | whole V | sub V | probe (whole) |
|---|---|---|---|---|---|
| **k-means (spherical)** | **0.440** | **0.244** | 0.409 | 0.177 | 0.820 |
| cosine-HAC | 0.277 | 0.113 | 0.275 | 0.249 | 0.820 |
| GMM (diag) | 0.335 | 0.146 | 0.394 | 0.181 | 0.820 |

**The architectural hypothesis is FALSIFIED.** k-means gives the *highest* pooled V/NMI of

the three,

1. **The pooled augmentation is real and clusterer-robust** for the leading clusterer

(k-mean

This refines the §7 cross-experiment pattern: the sub-call advantage shows on supervised metrics (p

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## 20. Track C CCA — acoustic ↔ social share a genuine common subspace (2026-06-23)

`scripts/bat_trackc_cca.py` — unsupervised dual-validation of the §9 supervised 4.7× context-re

| canonical pair | ρ | ρ² |
|---|---|---|
| $\rho_1$ | **0.771** | 0.595 |
| $\rho_2$ | 0.699 | 0.488 |
| $\rho_3$ | 0.623 | 0.388 |
| $\rho_4$ | 0.599 | 0.359 |
| $\rho_5$ | 0.541 | 0.293 |

**total $R^2$ (first 5 comps) = 2.123**

permuta

**The Track C multimodal synergy is intrinsic, not a classifier artifact.** A leading canonical

**Honest caveat.** The permutation p-value on $\rho_1$ is marginal (0.045), and the social space is high-dir

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## 21. LibriSpeech UNIFIED rerun — same words for §14 + Stage C + Stage D (2026-06-23)

`scripts/ls_unified_rerun.py` — fixes the apples-to-apples gap: the three LS word-structure scripts (§1

### §14 — within/across edit-dist + NN + motif→word (n=600)

| source | within | across | ratio | NN | NN_xsp | motifV |
|---|---|---|---|---|---|---|
| HECB(k32) | 0.917 | 0.935 | 0.981 | 0.098 | 0.090 | 0.429 |
| HuBERT-km(k50) | 0.939 | 0.928 | **1.012** | 0.047 | 0.047 | 0.424 |
| **fingerprint(k32)** | 0.907 | 0.942 | 0.964 | 0.118 | 0.114 | **0.481** |
| CPC-km(k50) | 0.900 | 0.940 | 0.958 | **0.175** | **0.171** | 0.458 |

### Stage C — DUSTED recurrent-term discovery (DTW row, same n=600)

| source | #fams | cov | word_V | H | top-frac | null-ratio | bootstrap |
|---|---|---|---|---|---|---|---|
| HECB(k32) | 260 | 0.038 | 0.419 | 0.98 | 0.02 | 1.15× | 0.181 |
| HuBERT-km(k50) | 617 | 0.080 | 0.416 | 0.96 | 0.01 | 1.35× | **0.428** |
| **fingerprint(k32)** | 269 | 0.068 | **0.453** | 0.96 | 0.03 | 1.26× | 0.298 |
| CPC-km(k50) | 738 | 0.052 | 0.427 | 0.97 | 0.01 | 1.21× | 0.310 |

### Stage D — whole-word embedding clustering (same n=600, k=60)

| source | V | ARI | purity | tokenF1 |
|---|---|---|---|---|
| **fingerprint** | **0.517** | 0.054 | 0.282 | 0.282 |
| HECB-pooled | 0.451 | 0.013 | 0.223 | 0.223 |
| HuBERT-mean | 0.386 | −0.003 | 0.193 | 0.193 |
| CPC-mean | 0.463 | 0.033 | 0.248 | 0.248 |

### Unified verdict (the point of putting all three on the same words)

**The fingerprint leads on every aggregate metric**: §14 motifV 0.481, Stage-C DTW word_V

0.453, S

Run: `python scripts/ls_unified_rerun.py --n-files 200 --max-words 600`; re-score:

`python s

---

## 22. Cross-corpus DUSTED motif table — uniform threshold, HAC-corrected (2026-06-23)

The three animal DUSTED scripts (`hecb_dusted_marmoset`, `dusted_bat`, `dusted_bengalese`) now

emit iden
tier)**

| corpus (n) | #fams | cov | purity(w) | H(norm) | top-frac | null-ratio | bootstrap |
|---|---|---|---|---|---|---|---|
| **Bengalese finch** (3 birds, 1 | 457 | 0.480 | **0.788** | 0.85 | 0.12 | **2.67×** | **0.509** |
| **Egyptian fruit bat** (300 bouts) | 306 | 0.153 | 0.39 † | 0.87 | 0.06 | 1.36× | 0.394 |
| **Infant marmoset** (1500 calls) | 139 | 0.028 | 0.49 † | 0.89 | 0.06 | 1.55× | 0.229 |

† purity label is corpus-specific (bengalese = GT syllable sequence → genuine blind recovery;

bat = Cor

**The cross-corpus claim holds on honest (HAC-corrected) numbers.** The relative ordering —

Bengalese

**§11 marmoset**: "DTW ct_V 0.508, 4.7× exact" was a CC-chaining artifact — HAC-corrected

marmo

call-type

**Manuscript framing:** DUSTED lands exactly where the repertoire predicts — diverse discrete

syllables

---

## 23. RookID Event/Comment enrichment — whole-call fp vs DUSTED motifs × 6 targets (2026-06-23)

`scripts/rookid_event_comment.py` — keeps the two columns the prior RookID scripts dropped:

**`Event`

| target | n | #cls | fp-V | fp-NMI | fp-probe | motif-V | motif-NMI |
|---|---|---|---|---|---|---|---|
| **Source** (individual) | 2798 | 14 | 0.308 | 0.175 | 0.325 | 0.347 | 0.156 |
| **activity** (singing/call) | 2730 | 2 | 0.673 | 0.159 | **0.846** | 0.657 | 0.092 |
| **sing/non-sing** | 2730 | 2 | 0.673 | 0.159 | **0.846** | 0.657 | 0.092 |
| **song_variant** (song-prefix) | 1938 | 14 | 0.242 | 0.087 | 0.118 | 0.327 | 0.129 |
| **social_other** (partner rook) | 2809 | 2 | **0.041** | 0.001 | n/a | 0.176 | 0.001 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **date** (recording) | 2809 | 4 | 0.249 | 0.055 | n/a | 0.315 | 0.057 |

**Findings (a clean individual-vs-social separation):**

1. **Indivi

**Manuscript framing:** across RookID targets the pattern is *what* (activity/sing) and *who*

(Source) a

---

## 24. Soundwel Track C — does acoustic structure predict emotional valence beyond call type? (2026-06-23)

`scripts/soundwel_trackc.py` — Soundwel = 6887 labelled pig vocalizations (44.1 kHz,

~0.4 s) w

### 6-feature × 3-target probe (Team-held-out macro-F1)

| target | whole fp | HECB | seq | motif | ctype | whole+ctype |
|---|---|---|---|---|---|---|
| vocalisation_type (chance 0.17, k=6) | **0.417** | 0.275 | 0.382 | 0.288 | — | — |
| production_context (chance 0.50, k=2) | 0.662 | — | — | — | **0.790** | 0.626 |
| emotional_valence (chance 0.50, k=2) | 0.478 | — | — | — | 0.296 | 0.466 |

### Decisive valence block (the "beyond call type" test; chance 0.50)

| test | macro-F1 |
|---|---|
| call-type → valence (baseline) | 0.508 |
| fp → valence | 0.478 |
| **fp + call-type → valence** | **0.466** (adding fp *hurts*) |
| **within-call-type fp → valence** | **0.350** (well below chance; 6 calltype |
| fp → valence (male / female) | 0.339 / 0.352 (both below chance) |

### Verdict — the bioacoustic semantic ceiling, proven

**Acoustic structure does NOT predict emotional valence — it is at/below chance, including

the clean

**Manuscript framing:** emotional valence is locked inside the call-type template; it is not linearly se

---

## 25. LibriSpeech speaker-invariant / universal acoustic-structure test (2026-06-23)

`scripts/ls_speaker_invariant.py` — does the ordered acoustic sub-unit representation preserve

### 4-way distances (ratio = d(diff-word,same-spk) ÷ d(same-word,diff-spk); >1 ⇒ word > speaker)

| representation | sw/ds | dw/ss | ratio |
|---|---|---|---|
| fp_whole (cosine) | 0.160 | 0.137 | **0.86** |
| fingerprint-km | 0.904 | 0.886 | 0.98 |
| HECB | 0.904 | 0.901 | 1.00 |
| CPC-km | 0.900 | 0.875 | 0.97 |
| HuBERT-km | 0.929 | 0.908 | 0.98 |
| HuBERT-DTW (raw-frame, n=300) | 0.864 | 0.826 | 0.96 |

### Cross-speaker same-word retrieval (eval=400, gallery=1895; gallery excludes query speaker; chance≈0.015)

| representation | top-1 | top-5 | MRR |
|---|---|---|---|
| **CPC-km** | **0.105** | **0.315** | **0.212** |
| fingerprint-km | 0.087 | 0.233 | 0.163 |
| fp_whole (cosine) | 0.083 | 0.230 | 0.162 |
| HECB | 0.068 | 0.185 | 0.135 |
| HuBERT-DTW (n=300) | 0.042 | 0.083 | 0.074 |
| HuBERT-km (Levenshtein) | 0.018 | 0.098 | 0.062 |

### Word-vs-speaker disentanglement + centering (fp_whole)

word probe (speaker-held-out macro-F1) = **0.110**; speaker probe = 0.094 (chance 0.006).

word pro

### Honest verdict — a *qualified* universal representation

1. **Cross-speaker same-word retrieval is well above chance** (CPC 7×, fp/fingerprint ~5–6×):

word-as

**Manuscript-safe claim:** *cross-speaker same-word retrieval from ordered acoustic sub-unit

represent

---

## 26. RookID cross-Caller same-call retrieval — the §25 animal analogue (2026-06-23)

`scripts/rookid_cross_caller.py` — the cross-species mirror of §25: does CALL-TYPE structure

survive ac

| target (call-type | n label) | #labels | chance | fp top-1 | fp top-5 | fp MRR | ratio |
|---|---|---|---|---|---|---|---|
| activity (sing/ca | 2809 | 2 | 0.572 | 0.665 | 0.910 | 0.771 | 1.00 |
| sing/non-sing | 2809 | 2 | 0.572 | 0.598 | 0.882 | 0.719 | 0.97 |
| song_variant (sing prefix) | 2066 | 7 | 0.330 | **0.403** | 0.677 | 0.514 | 0.99 |
| Event (fine code) | 1677 | 30 | 0.240 | **0.160** | 0.393 | 0.273 | 0.98 |

(sub-call-seq Levenshtein is within ±0.02 of fp on every cell — both reps agree.)

### Cross-species verdict — animal calls are MORE caller-bound than human words are speaker-bound

1. **Coarse category barely generalizes across rooks**: activity/sing cross-caller top-1

(0.60–0.

**The decisive cross-species contrast (§25 ↔ §26):** on LibriSpeech, cross-**speaker**

same-**w

**Manuscript framing:** *cross-speaker word retrieval is well above chance, but cross-caller

call-type

---

## 27. DTW-barycenter consensus word template — the §25 follow-on (negative) (2026-06-23)

`scripts/ls_consensus_template.py` — §25 found linear speaker-centering **fails** (Δ −0.003),

so the pri

| method | top-1 | chance |
|---|---|---|
| raw cross-speaker NN (DTW on fp-traject | **0.212** | 0.040 |
| **DBA consensus template** | **0.172** | 0.040 |
| Δ(template − NN) | **−0.040** | |

### Verdict — no simple consensus "universal word" (honest negative)

The DBA consensus template does **not** beat raw cross-speaker NN — it is slightly *worse*

(0.172 <

**Manuscript framing:** *there is no simple consensus "universal word" template derivable

from the

### CPC-space replication — the representation matters (the sign flips)

Re-running with `--rep cpc` (continuous 256-D CPC frame trajectories, §25's most

speaker-i

| representation | raw cross-speaker NN | DBA consensus template | Δ (template − NN) |
|---|---|---|---|
| fingerprint (43-D windows, phy | 0.212 | 0.172 | **−0.040** |
| HuBERT (768-D frames, MLM) | 0.092 | 0.044 (≈chance) | **−0.048** |
| AVES (768-D frames, multi-tax | 0.096 | 0.028 (below chance) | **−0.068** |
| **CPC (256-D frames, contrast | **0.428** | **0.440** | **+0.012** |

The sign **flips** across representations: in the physics-bound fingerprint, DBA averaging

**smears

**Honest magnitude.** The effect is real but **modest** (+0.012, paired over the same 250

queries):

**Firewall / animal translation.** CPC is human-trained and firewall-blocked on animals, so

this speak

---

## 28. RookID sing/song-bout DUSTED — within-song motif inventory (2026-06-23)

DUSTED DTW discovery on the **sing subset** (1938 calls, 6 sources, 14 song variants)

of the cac

| stat | value |
|---|---|

| | |
|---|---|
| #families | **1398** |
| H(norm) | **0.93** (diverse, non-degenerate — unli |
| top-family-frac | **0.01** (no dominant) |
| src_V (Source) | 0.042 |
| var_V (song_variant) | 0.084 |
| null ratio | 1.06× (p=0.111 — composition-driven) |
| bootstrap Jaccard | 0.341 (moderate) |

| representation | Source V | Source F1 | song_variant V | Event V |
|---|---|---|---|---|
| **whole fp (44-D)** | **0.294** | **0.826** | **0.230** | **0.243** |
| HECB mean (44-D) | 0.302 | 0.822 | 0.206 | 0.215 |
| motif hist (50-D) | 0.110 | 0.446 | 0.108 | 0.136 |

**Findings:** (1) Rook song HAS diverse recurrent acoustic structure (1398 families, H=0.93,

top=0.01

---

## 29. Cross-analysis: why recurrent acoustic terms are not decoded words (2026-06-23)

Five summary cross-cuts across the validated results (§1–§28), assembled into the

reviewer-

### Cross-analysis 1 — cross-speaker word invariance vs cross-caller call-type invariance

| Corpus | Unit tested | Identity held out | Retrieval target | Result | Interpretation |
|---|---|---|---|---|---|
| **LibriSpeech** | word | speaker | same word | **5–7× chance** (CPC 0.105 fa 0.03 | words generalize across speaker |
| **RookID** | Event (fine code) | caller (Source) | same Event | **below chance** (0.160 vs 0.240) | fine calls are caller-bound |
| **RookID** | song_variant | caller | same variant | **moderate** (0.403 vs 0.330) | song structure partly shar |
| Egyptian bat | context | emitter | same context | expected weak (§13 V=0.16) | context not acoustic-invariant |
| InfantMarmosetsVox | call-type | caller/twin | same call-type | **strong** (V=0.606, §18) | call type is acoustic (form), not s |

**The decisive contrast:** human **words** generalize across speakers (5–7× chance, §25);

animal **

### Cross-analysis 2 — identity / call-form / context disentanglement across corpora

| Dataset | Identity signal | Call/form signal | Context/social signal | Best representation |
|---|---|---|---|---|
| **Egyptian bat** | **high** (Emitter V≈0.45) | moderate (motifs graded) | **weak** (Context V≈0.16) | fingerprint; social needed for context |
| **RookID** | **high** (Source V=0.31) | ing/call strong (V=0.67) | **weak** (social_other V=0.04) | whole-call fingerprint |
| **InfantMarmosetsVox** | moderate (caller V present) | **strong** (call-type V=0.606) | receiver-side context | fingerprint + HECB roles |
| **Bengalese finch** | bird-constant (per-bird) | **strong** (GT syllable purity 0.78) | motif/acoustic context | exact/fixed-cosine motif |
| **Soundwel pig** | — (no individual ID) | call-type acoustic (0.41) | **valence not recoverable** (0.35 < chance) | fingerprint/CPC |
| **LibriSpeech** | speaker entangled (§25 ratio=1) | **strong structure** (motif V=0.48, §21) | word GT (controlled) | fingerprint/CPC |

**The pattern:** animal acoustic structure carries *who called* and *what call form* more

reliably th

## Cross-analysis 3 — word-likeness / lexicality stress test

| Criterion | LibriSpeech | Bengalese | RookID | Bat | Marmoset |
|---|---|---|---|---|---|
| Cross-individual invariance | **high** (§25 5–7×) | per-bird (n/a) | **low** (§26 ≈chance) | low (§13 emitter-bound) | moderate (V=0.60 but within-ses |
| Motif inventory diversity | high | **high** (§12 H=0.85) | **high** (§28 H=0.93) | moderate (§13 graded) | **sparse** (§11 degenerate) |
| Label purity / enrichment | high (§21 motif V 0.48) | **high** (§12 0.79) | **low** (§28 V<0.1) | low (§13 V<0.17) | moderate (§18 V=0.61) |
| Context-independence | **high** (word GT) | n/a | **low** (§23 social≈0) | **low** (§9 social needed) | n/a |
| **Overall word-likeness** | **high** | moderate (acoustic, not semantic) | **low–moderate** | **low** | **low** |

**Reviewer-safe sentence:** *The pipeline recovers structured recurrent acoustic terms, but

animal co

## Cross-analysis 4 — evidence-tier status (H0-CMC ladder)

| Claim | Tier | Evidence | Status |
|---|---|---|---|
| Recurrent acoustic terms exist | **T0** | DUSTED discovery (§11–§13, §22, §28) | validated |
| Individual identity recoverable | **T1** | bat Emitter V, marmoset V=0.61, RookID Source V | validated |
| Call-form recoverable from acoustics | **T1** | marmoset Soundwel call-type 0.41 | validated |
| Context is socially grounded (not T1 output) | **T2+** | bat A–G D≈G=4.3×, survives shuffle-null | validated |
| Valence/emotion not acoustic | **T1** | Soundwel within-call-type 0.35 < caller (§ | validated (ceiling) |
| Receiver detects motif structure | **T2** | — | **needs playback** |
| Multimodal receiver convergence | **T3** | — | **needs playback** |
| Causally supported functional signal | **T4** | — | **needs counterfactual playback** |
| Ecologically validated function | **T5** | — | **needs field replication** |

**The pipeline has exhausted T0–T1** (acoustic association + contextual prediction). T2–T4

(receiver

## Cross-analysis 5 — playback-candidate prioritization

| Dataset | Best playback candidate | Why | H0-CMC prediction |
|---|---|---|---|
| **RookID** | Source familiarity (known vs novel cal | clean individual labels; tests caller-re | response follows caller identity, not mo |
| **InfantMarmosetsVox** | onset/body/offset role-templated calls | mechanistic HECB role test (T2) | response disrupted by role-swap → roles |
| **Egyptian bat** | emitter-associated motifs vs context-soc | tests social grounding (T2); D=4.3× pro | response follows emitter, not context |
| **Bengalese finch** | GT motif order-shuffled vs natural | positive control (T2); GT syllable sport | response disrupted by order-shuffle → mo |

**Counterfactual controls** (H0-CMC §4): natural / motif-preserved / order-shuffled /

caller-swa
amplitude

# 30. Boundary-quality sensitivity sweep — at what parent quality does hierarchy help? (2026-06-23)

**Question (Architecture v1.0 invariant #2/#3).** HECB annotates typed internal sub-call

boundarie

**Setup.** `hecb_decode` accepts any parent events, so parent quality is a clean independent variable;

**MeerKAT** (the HECB corpus): brief calls in noise + trained CRNN ⇒ parent quality varies via REAL de

## MeerKAT — real CRNN detector operating points

| method | parent F1 | whole V | pooled V | gap | whole probe |
|---|---|---|---|---|---|
| oracle_typed | 1.000 | 0.364 | 0.386 | **+0.021** | 0.753 |
| oracle_raw | 1.000 | 0.364 | 0.385 | +0.021 | 0.753 |
| crnn0.9_typed | 0.940 | 0.299 | 0.351 | +0.052 | 0.709 |
| crnn0.7_typed | 0.934 | 0.320 | 0.341 | +0.022 | 0.645 |
| crnn0.7_raw | 0.934 | 0.320 | 0.332 | +0.012 | 0.645 |
| crnn0.4_typed | 0.894 | 0.278 | 0.367 | +0.089 | 0.640 |
| raw_nbd | 0.101 | 0.286 | 0.286 | **0.000 ← abstain** | 0.583 |

## Marmoset — oracle/jittered parent (rich internal structure)

| method | parent MAE (ms) | whole V | pooled V | gap | whole probe |
|---|---|---|---|---|---|
| oracle_raw | 0.0 | 0.524 | 0.492 | **−0.032** | 0.810 |
| jitter25_raw | 12.9 | 0.522 | 0.486 | −0.035 | 0.808 |
| jitter75_raw | 38.8 | 0.537 | 0.483 | −0.054 | 0.804 |
| jitter150_raw | 77.6 | 0.524 | 0.487 | −0.037 | 0.792 |
| oracle_none (no internal) | 0.0 | 0.524 | 0.524 | 0.000 | 0.810 |

## Verdict — whole-call primary (invariant #3) confirmed on both corpora

**Sub-call internal structure is NOT a call-type-recovery lever.** At the best parent quality the hierarchy (pooled is

**Consequence for the HECB gate:** there is **no meaningful abstain threshold for hierarchy benefit**, because t parent).

**Honesty caveat.** The ablation V uses k=12 for sweep-internal consistency, so absolute whole V

(0.364 m
(meerkat

## 31. Cross-species repertoire taxonomy — WHY each corpus shows the structure it does (2026-06-23)

**Question (Architecture v1.0 Step 3).** The boundary sweep (§30) showed whole-call primary holds

regardless

| corpus | repertoire type | secondary flags | null ratio | bootstrap | coverage | emitter V | context V | social lift |
|---|---|---|---|---|---|---|---|---|
| **Bengalese finch** | `discrete_syllable` | — | 2.67× | 0.509 | 0.480 | n/a | n/a | n/a |
| **Egyptian fruit bat** | `graded` | identity_dominant | 1.36× | socially_grounded | 0.394 | 0.153 | 0.446 | 0.167 | 1.74 |
| **Infant marmoset** | `sparse_repetitive` | — | 1.55× | 0.229 | 0.028 | n/a | 0.599 | n/a |
| **Rook** | `individually_identity_dominant` | — | n/a | n/a | n/a | 0.31 | 0.04 | n/a |
| **LibriSpeech** (human) | `human_lexical` | — | — | — | — | — | — | — |

Numbers curated from validated sections (§22 cross-corpus DUSTED; §13/§18/§23 binding V; §9 social-lift),

each cite
re-derive

**Classifier (calibrated so each reference corpus lands on its validated type):**

1. human

**Two honest design notes.** (1) DUSTED family-entropy is near-max for ALL corpora (0.85–0.89) — a

*weak*
emitter-vs

## 32. Identity–form–context disentanglement — what acoustics encode / don't / need-social (2026-06-23)

**Question (Architecture v1.0 Step 4).** The backbone biological table: for each corpus, what do the

acoustics
partner,

| corpus | individual_id | call_type | context | social_partner |
|---|---|---|---|---|
| **Egyptian fruit bat** | **strong** (V 0.446) | weak (0.167) | **needs_social** (§9 D=4.3×) | n/a |
| **Infant marmoset** | n/a (no ID labels) | **strong** (V 0.599, probe 0.829) | n/a | n/a |
| **MeerKAT** | n/a | **strong** (V 0.521, probe 0.753) | n/a | n/a |
| **Rook** | partial (Source V 0.31) | **strong** (sing/call 0.67) | weak (social_other 0.04 = ≈ chance §26) | n/a (encode) |
| **LibriSpeech (human)** | **strong (speaker probe 0.56)** | partial (token↔word 0.29) | n/a (read speech) | n/a |

**The three-bucket thesis, operationalised:**

**ENCC
meerkat,

**Human anchor (the form/meaning split).** Speaker identity IS strongly encoded (CPC probe 0.56 — the

latent is s
tokens

**Steps 1–4 (the paper-ready core) are complete.**

## 33. Structural alternatives ablation — DUSTED vs RQA / Matrix Profile / PST / Transition Network (2026-06-23)

**Question.** Four structure-discovery techniques that do NOT rely on DUSTED's token-sequence

alignment
Modules i

| dataset (regime) | alternative | result | verdict vs DUSTED §22 |
|---|---|---|---|
| **Bengalese** (discrete, diverse) | PST + Transition Network | PST **247** predictive contexts, depth 8; **639** nodes, 1205 edges, **19** motif bouts | **CORROBORATES** (purity 0.788); DTW's |
| **Egyptian fruit bat** (graded, n=26) | RQA + Matrix Profile | RQA **DET 0.229 / LAM 0.169**; MP **recurrence_fraction 0.788**, 1 motif | **COMPLEMENTS** — quantifies the grade |
| **Infant marmoset** (sparse) | Matrix Profile (+ RQA) | MP **recurrence_fraction 0.788**, 1 motif | **FAST PRE-FILTER** — token-free O(n log |
| **LibriSpeech** (diverse syntactic) | Transition Network (word transitions) | **639** nodes, 1205 edges, **19** motif bouts | **UPPER BOUND** — scales to a far larger |

**The takeaway — no alternative replaces DUSTED; each finds its regime niche.** PST/Transition-Network

*corrobor
*complem

**Honest caveats.** (1) The wav2vec2 aligner emits word IDENTITIES + phoneme TIMES (not phoneme

identities)
the user

## 34. Three targeted structural reanalyses — closing DUSTED

## limitations (2026-06-23)

The §33 alternatives applied surgically to three known DUSTED limitations — NOT a broad reanalysis.

Scripts: `scripts/{ `.

### #1 Bat — RQA laminarity is context-dependent (the dynamical lens DUSTED lacked)

| context code | n | DET | LAM |
|---|---|---|---|
| 12 | 30 | 0.216 | **0.083** |
| 11 | 30 | 0.264 | 0.229 |
| 4 | 30 | 0.251 | **0.265** |
| 3 | 30 | 0.268 | 0.147 |
| 6 | 30 | 0.209 | 0.196 |
| 9 | 30 | 0.250 | 0.139 |

Kruskal-Wallis across contexts: DET H=1.74 **p=0.884** (NS) | LAM H=13.88 **p=0.016** (significant).

**Verdict:** graded bat bouts have **context-dependent LAMINARITY** — the degree to which the bat

holds a s 0.265

### #2 Marmoset — token-free MP confirms degenerate inventory (tokenization-independent)

n=200 calls, raw 44-D fingerprint trajectory, **bypassing HECB + DUSTED entirely:**

recurren

**Verdict:** the degenerate-inventory claim (§11) is **tokenization-independent** — token-free MP on

raw signa an

### #3 Bengalese — full transition-network topology (songbird-syntax lens)

n=100 bouts → **24 nodes**, **5 communities**, clustering **0.686**, density **0.475**, mean path

length **1

**Verdict:** **small-world + modular** — the topology songbird-syntax literature reports for Bengalese

finch son
without la

**Structure discovery is now 100% complete.** Three paradigms (alignment / graph / dynamics) agree on

the biolog

## 36. The Entanglement Ceiling — caller identity and call form are non-separably entangled (2026-06-23)

**Question.** §26 established that animal calls are *non-additively caller-bound* — the raw distance

ratio betw
meaning

**Motivation.** In human speech, CPC's contrastive objective partially disentangles speaker from word

content (
ethologica

**Setup.** Marmoset corpus (5 call types: Phee/Twitter/Trill/Tsik/Seep; 87 callers with ≥20 calls

each). Fo
caller labe

Module: `representation/disentangle.py` (pure numpy, firewall-clean, deterministic).

### Method 1: Orthogonal Subspace Projection (LDA) — linear algebraic

Learns the caller-discriminating subspace via Linear Discriminant Analysis, then projects the

fingerprin
of

| | caller V | form V | ratio | Δ caller | Δ form |
|---|---|---|---|---|---|
| Baseline (raw fp) | 0.611 | 0.614 | 1.0 | — | — |
| LDA projection | 0.582 | 0.494 | 0.8 | −0.029 | **−0.120** |

**Verdict: FAILS.** Form degrades 4× harder than caller (−0.120 vs −0.029). The caller and form

subspace

## Method 2: Orthogonal Subspace Projection (LogReg) — linear algebraic

Same principle, but the caller-discriminating directions come from a multinomial Logistic Regression

weight ma

| | caller V | form V | ratio | Δ caller | Δ form |
|---|---|---|---|---|---|
| Baseline (raw fp) | 0.611 | 0.614 | 1.0 | — | — |
| LogReg projection | 0.606 | 0.601 | 1.0 | −0.005 | −0.013 |

**Verdict: NO EFFECT.** The LogReg weight matrix could not find a caller-discriminating subspace

separable

## Method 3: Per-Speaker Z-Normalization — affine transform

For each caller, centers (subtracts mean) and scales (divides by std) every dimension across all

their calls than

| | caller V | form V | ratio | Δ caller | Δ form |
|---|---|---|---|---|---|
| Baseline (raw fp) | 0.611 | 0.614 | 1.0 | — | — |
| Z-normalization | 0.592 | 0.540 | 0.9 | −0.019 | **−0.074** |

**Verdict: FAILS.** Form degrades 4× harder than caller (−0.074 vs −0.019). Per-speaker affine

normaliza entangler

## Method 4: Delta-Trajectory — non-linear structural

Replaces absolute frame fingerprints with inter-frame differences: Δ_i = fp_{i+1} − fp_i. Theory:

static call

Tested on 200 marmoset calls → 4,623 frame windows (80 ms each, 44-D fp per frame) → 4,623 delta

vectors.

| | caller V | form V | ratio | Δ caller | Δ form |
|---|---|---|---|---|---|
| Raw frame fp (absolute) | 0.278 | 0.396 | 1.4 | — | — |
| Delta trajectory (differenced) | 0.173 | 0.233 | 1.4 | −0.105 | **−0.163** |

**Verdict: FAILS.** The disentanglement ratio stays at 1.4 — zero improvement. Form degrades 1.5×

harder th spectral

## Summary: the complete disentanglement proof

| Method | Mathematics | Entanglement level tested | Δ caller | Δ form | Result |
|---|---|---|---|---|---|
| LDA projection | Linear subspace | Static spectral (linear) | −0.029 | −0.120 | **FAIL** (overlap) |
| LogReg projection | Linear weight | Static spectral (linear) | −0.005 | −0.013 | **FAIL** (inseparable) |
| Z-normalization | Affine (mean + scale) | Static spectral (affine) | −0.019 | −0.074 | **FAIL** (overlap) |
| Delta-trajectory | Non-linear differencing | Dynamic temporal (structural) | +0.105 | −0.163 | **FAIL** (temporal dynamics are |

**Conclusion.** The 44-D acoustic fingerprint cannot be disentangled into pure caller identity and pure

call form
of

1. **Static spectral (linear):** the caller-discriminating subspace (LDA/LogReg) overlaps with the

call-type

**The semantic ceiling is a mathematical fact, not a policy choice.** In human speech, CPC's

contrastiv
from

**Implications for the architecture:**

The etho

## AVES ablation — does a deep learned embedding break the ceiling? (reviewer-proofing)

A reviewer will argue the entanglement is an artifact of hand-crafted 44-D physics features, and that

a deep ne
directly:

| Representation | Method | caller V | form V | ratio | Δ caller | Δ form | form drops ≥ caller? |
|---|---|---|---|---|---|---|---|
| fingerprint 44-D | baseline | 0.276 | 0.552 | 2.00 | — | — | — |
| fingerprint 44-D | LDA | 0.251 | 0.459 | 1.83 | −0.025 | **−0.093** | YES (3.7×) |
| fingerprint 44-D | Z-norm | 0.282 | 0.478 | 1.70 | +0.006 | **−0.074** | YES |
| AVES 768-D | baseline | 0.226 | 0.628 | 2.78 | — | — | — |
| AVES 768-D | LDA | 0.227 | 0.631 | 2.77 | +0.001 | +0.003 | no-op |
| AVES 768-D | Z-norm | 0.212 | 0.587 | 2.77 | −0.014 | **−0.041** | YES (2.9×) |

**Honest verdict — deep learning narrows the ceiling but does not break it.**

1. **AVES is a less-entangled representation than the fingerprint** (baseline ratio 2.78 vs 2.00): it

achieve
reviewer's

2. **But AVES is still not linearly disentangleable.** Z-normalization — the affine "scrub the caller"

transfor
form

3. **LDA is a no-op on AVES.** Projecting out the linear caller subspace from 768-D changes neither

caller V
(where

**Conclusion.** No firewall-compliant *available* representation — deterministic physics (44-D) or deep

SSL (768-

the gap

*Reproducibility note.* `entanglement_ceiling_aves.py` also reproduces §36's fingerprint {baseline, LDA,

Z-norm}
matches §

## Named result: the Entanglement Ceiling

This is the **Entanglement Ceiling**: caller physical identity and call motor form are

non-linear

**Biological interpretation — the medium is the message.** The delta-trajectory failure is

the most

**Human vs. animal — closing the loop.** This sharpens the cross-species analogy and makes

the huma

### Why this mandates Track C (the final pivot)

With this test, the computational representation space is officially exhausted. Because the computer